

Abinit workshop 2026

In search of the electron-phonon contribution to total energy

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In most *ab-initio* code, the total energy is:

- computed at fixed atomic positions (BO)
- with vibrational free energy (phonon energy - QHA) $\propto 1/\sqrt{M}$
- with electronic free energy
- with third/fourth-order anharmonic corrections (e.g. TDEP, SSCHA, ALAMODE, hiphive) $\propto 1/M$

Goal: to explore “forgotten” contributions to total energy from the Born-Oppenheimer expansion.

Born-Oppenheimer & adiabatic approximation

$$\hat{\tilde{H}}\{\mathbf{R}\} |\psi_i\{\mathbf{R}\}\rangle = \mathcal{E}_i |\psi_i\{\mathbf{R}\}\rangle$$

$$\hat{\tilde{H}}\{\mathbf{R}\} \equiv \tilde{T}^N + \hat{H}^{\text{BO}}\{\mathbf{R}\}$$

$$\hat{H}^{\text{BO}}\{\mathbf{R}\} |\phi_j\{\mathbf{R}\}\rangle = E_j^{\text{BO}}\{\mathbf{R}\} |\phi_j\{\mathbf{R}\}\rangle$$

$$\hat{H}^{\text{BO}}\{\mathbf{R}\} \equiv \hat{H}^e\{\mathbf{R}\} + V^{\text{NN}}\{\mathbf{R}\}$$

Exact link through the nuclear dynamics:

$$|\psi_i\{\mathbf{R}\}\rangle = \sum_j \chi_{ij}\{\mathbf{R}\} |\phi_j\{\mathbf{R}\}\rangle$$

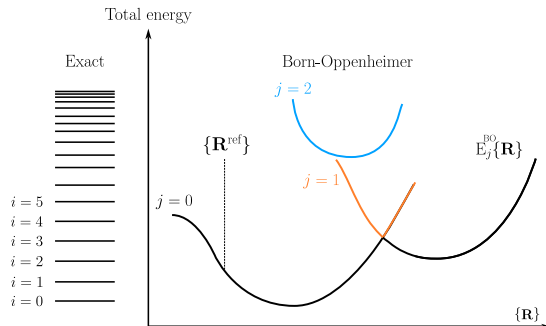
The *scalar adiabatic approximation* is:

$$|\psi_i\{\mathbf{R}\}\rangle \approx \chi_{ij}^{\text{sca}}\{\mathbf{R}\} |\phi_j\{\mathbf{R}\}\rangle$$

$$(\tilde{T}^N + E_j^{\text{BO}}\{\mathbf{R}\}) \chi_{ij}^{\text{sca}}\{\mathbf{R}\} = \mathcal{E}_{ij}^{\text{sca}} \chi_{ij}^{\text{sca}}\{\mathbf{R}\}.$$

(other approximation exists: optimal & hybrid)

$|\phi_j\{\mathbf{R}\}\rangle$ form a complete orthonormal basis $\forall \{\mathbf{R}\}$



M. Born and R. Oppenheimer, Ann. der Physik **389**, 457 (1927)

Mass-scaling analysis

We choose a local BO minimum as the reference configuration $\{\mathbf{R}_j^0\}$ and Taylor expand the mass in λ .

We choose a characteristic mass M_0 : $m_\kappa \equiv M_\kappa/M_0$.

Define a small parameter λ : $M_\kappa(\lambda) = m_\kappa \lambda^{-4}$

Physical mass obtained for $\lambda_0 \equiv M_0^{-1/4}$ [at most 0.16 (for H)]

$$\tilde{\mathcal{H}}_\lambda\{\mathbf{R}\} |\psi_{i\lambda}\{\mathbf{R}\}\rangle = \mathcal{E}_{i\lambda} |\psi_{i\lambda}\{\mathbf{R}\}\rangle$$

$$\tilde{\mathcal{H}}_\lambda\{\mathbf{R}\} = \sum_{\kappa\alpha} \frac{\lambda^4}{2m_\kappa} \frac{\partial^2}{\partial R_{\kappa\alpha}^2} + \hat{H}^{\text{BO}}\{\mathbf{R}\}$$

$$X(\lambda) = X^{(0)} + \lambda X^{(1)} + \lambda^2 X^{(2)} + \mathcal{O}(\lambda^3),$$

$$X^{(n)} = \frac{1}{n!} \frac{\partial^n X}{\partial \lambda^n} \Big|_{\lambda=0}.$$

The physical energy is obtained with

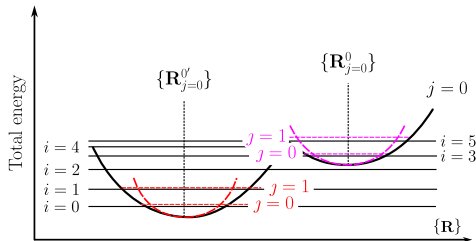
$$\mathcal{E}_{i\lambda_0} = \mathcal{E}_i^{(0)} + \lambda_0 \mathcal{E}_i^{(1)} + (\lambda_0)^2 \mathcal{E}_i^{(2)} + \mathcal{O}(\lambda_0^3),$$

where odd terms in λ_0 vanishes.

Local BO minimum $\rightarrow$ harmonic oscillator
nuclear displacement amplitude $\Delta\mathbf{R} \propto \frac{1}{\sqrt{M\omega}}$

with $\omega \propto \sqrt{K/M}$

$\Delta\mathbf{R} \propto M^{-1/4} \propto \lambda$



Mass-scaling analysis

Expansion of BO hypersurface in displacements around equilibrium

$$E_j^{\text{BO}}\{\mathbf{R}\} = E_{j|\mathbf{0}}^{\text{BO}} + E_j^{\text{BO}\{2\}} \cdot \Delta\mathbf{R}_j^2 + \mathcal{O}(\Delta\mathbf{R}_j^3).$$

We consider only $E_j^{\text{BO}\{2\}}$ with auxiliary harmonic oscillator (ho):

$$\check{\mathfrak{H}}_j^{\text{ho}}\{\Delta\mathbf{R}\} \equiv \sum_{\kappa\alpha} \frac{\check{P}_{\kappa\alpha}^2}{2m_\kappa} + E_j^{\text{BO}\{2\}} \cdot \Delta\mathbf{R}_j^2$$

$$\check{\mathfrak{H}}_j^{\text{ho}}\{\Delta\mathbf{R}\} \vartheta_{jj}\{\Delta\mathbf{R}\} = \mathfrak{E}_{jj}^{\text{ho}} \vartheta_{jj}\{\Delta\mathbf{R}\},$$

The

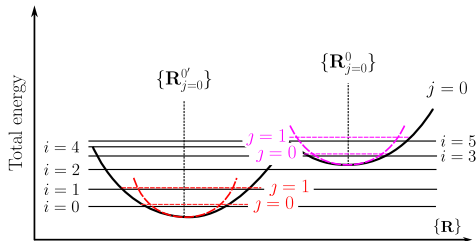
$$|\Psi_{i\lambda}\{\mathbf{R}\}\rangle \equiv N_\lambda \vartheta_{jj}\{\Delta\mathbf{R}_j/\lambda\} |\phi_j\{\mathbf{R}\}\rangle$$

$$\mathcal{E}_{i\lambda} = E_{j|\mathbf{0}}^{\text{BO}} + \lambda^2 \mathfrak{E}_{jj}^{\text{ho}} + \mathcal{O}(\lambda^3),$$

is solution (to 2nd order) of exact

$$\check{\mathcal{H}}_\lambda\{\mathbf{R}\} |\psi_{i\lambda}\{\mathbf{R}\}\rangle = \mathcal{E}_{i\lambda} |\psi_{i\lambda}\{\mathbf{R}\}\rangle$$

$\vartheta_{jj}\{\Delta\mathbf{R}\}$ is the nuclear wavefunction with nuclear state j for a j BO hypersurface



Rescaling of displacements and momenta

Once we have solved the aux. harmonic oscillator, we can obtain the full solution with a rescaling:

$$\rho_{\kappa\alpha j} \equiv \Delta R_{\kappa\alpha j} / \lambda$$

$$\tilde{\pi}_{\kappa\alpha} \equiv \lambda \frac{\partial^2}{\partial R_{\kappa\alpha}^2}.$$

$$\tilde{\mathcal{H}}_\lambda[\rho_j] |\psi_{i\lambda}[\rho_j]\rangle = \mathcal{E}_{i\lambda} |\psi_{i\lambda}[\rho_j]\rangle$$

$$|\psi_{i\lambda}[\rho_j]\rangle = \chi_{jj\lambda}[\rho_j] |\phi_j[\lambda\rho_j]\rangle$$

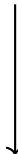
$$\tilde{\mathfrak{H}}_{j\lambda}[\rho_j] \chi_{jj\lambda}[\rho_j] = \mathfrak{E}_{jj\lambda} \chi_{jj\lambda}[\rho_j]$$

$$\tilde{\mathfrak{H}}_{j\lambda}[\rho_j] = \tilde{\mathfrak{H}}_j^{\text{ho}}[\rho_j] + \Delta\hat{\mathfrak{H}}_{j\lambda}[\rho_j]$$

$$\tilde{\mathfrak{H}}_j^{\text{ho}}[\rho_j] \equiv \sum_{\kappa\alpha} \frac{\tilde{\pi}_{\kappa\alpha}^2}{2m_\kappa} + E_j^{\text{BO}\{2\}} \cdot \rho_j^2, \quad \leftarrow \text{rescaled harmonic oscillator equation}$$

$$\begin{aligned} \Delta\tilde{\mathfrak{H}}_{j\lambda}[\rho_j] \equiv & \lambda E_j^{\text{BO}\{3\}} \cdot \rho_j^3 + \lambda^2 \left(\tilde{T}_j^{\text{vec}(1)\{1\}} \cdot \rho_j + E_j^{\text{BO}\{4\}} \cdot \rho_j^4 + E_j^{\text{Born}}[\lambda\rho_j] \right) + \lambda^4 \tilde{\Sigma}_{j\lambda}(\mathcal{E}_{i\lambda}) \\ & + \sum_{n=3} \lambda^n \left(\tilde{T}_j^{\text{vec}(1)\{n-1\}} \cdot \rho_j^{n-1} + E_j^{\text{BO}\{n+2\}} \cdot \rho_j^{n+2} \right) + \sum_{n=4} \lambda^n \left(\tilde{T}_j^{\text{vec}(2)\{n-2\}} \cdot \rho_j^{n-2} \right) \end{aligned}$$

Correction : anharmonic + others



Beyond the electronic (BO min.) + vibrational energy

To 4th order, 3 contributions.

Vector potential (velocity contribution):

$$\mathcal{E}_i^{(4),\text{vec}} = \langle \vartheta_{jj} | \check{T}_j^{\text{vec}(1)\{1\}} \cdot \check{\rho}_j | \vartheta_{jj} \rangle_{\rho_j}$$

Anharmonic contributions (also quasi-harmonic)

$$\begin{aligned} \mathcal{E}_i^{(4),\text{anh}} = & \langle \vartheta_{jj} | E_j^{\text{BO}\{4\}} \cdot \check{\rho}_j^4 | \vartheta_{jj} \rangle_{\rho_j} \\ & + \langle \vartheta_{jj} | (E_j^{\text{BO}\{3\}} \cdot \check{\rho}_j^3) \check{\mathfrak{G}}_{\perp jj}^{\text{ho}} (E_j^{\text{BO}\{3\}} \cdot \check{\rho}_j^3) | \vartheta_{jj} \rangle_{\rho_j} \end{aligned}$$

Born contribution (electron-phonon ?)

$$\mathcal{E}_i^{(4),\text{elph}} \equiv \langle \vartheta_{jj} | E_{j|\mathbf{0}}^{\text{Born}} | \vartheta_{jj} \rangle = \sum_{\kappa\alpha} \left\langle \frac{\partial \phi_j}{\partial R_{\kappa\alpha}} \Big|_{\mathbf{0}} \left| \frac{\hat{\mathcal{P}}_{\perp j}^{\text{BO}}[\mathbf{0}]}{2m_{\kappa}} \right| \frac{\partial \phi_j}{\partial R_{\kappa\alpha}} \Big|_{\mathbf{0}} \right\rangle$$

Energy	Equation	Mass-scaling
$\mathcal{E}_i^{(0)} = E_{j \mathbf{0}}^{\text{BO}}$	(44)	$\mathcal{O}(\lambda^0)$
$\mathcal{E}_i^{(2)} = \mathfrak{E}_{jj}^{\text{ho}}$	(46)	$\mathcal{O}(\lambda^2)$
$\mathcal{E}_i^{(4),\text{vec}}$	(81)	$\mathcal{O}(\lambda^4)$
$\mathcal{E}_i^{(4),\text{anh}}$	(78)	$\mathcal{O}(\lambda^4)$
$\mathcal{E}_i^{(4),\text{elph}}$	(88)	$\mathcal{O}(\lambda^4)$
$\mathcal{E}_i^{(6),\text{elph}}$	(S26)	$\mathcal{O}(\lambda^6)$
$\mathcal{E}_i^{(6),\text{elm}}$	(S27)	$\mathcal{O}(\lambda^6)$
$\mathcal{E}_i^{(6),\text{vec}}$	(S37)	$\mathcal{O}(\lambda^6)$
$\mathcal{E}_i^{(6),\text{anh}}$	(S39)	$\mathcal{O}(\lambda^6)$
$\mathcal{E}_i^{(6),\text{mix}}$	(S38)	$\mathcal{O}(\lambda^6)$

Electron-phonon contribution to total energy

We consider the ground-state ($i = 0$), for solids, using DFT, Slater determinant of Kohn-Sham wavefunctions

$$\tilde{\mathcal{H}}_\lambda[\rho_j] |\psi_\lambda[\rho_j]\rangle = \mathcal{E}_\lambda |\psi_\lambda[\rho_j]\rangle$$

$$\mathcal{E}_\lambda = E_0^{\text{BO}} + \lambda^2 \mathcal{E}^{(2)} + \lambda^4 \mathcal{E}^{(4)} + \mathcal{O}(\lambda^5),$$

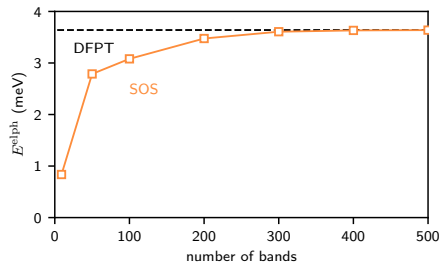
Physical total energy ($\lambda = \lambda_0$):

$$E^{\text{ph}} \equiv \lambda_0^2 \mathcal{E}^{(2)} = \lambda_0^2 \mathfrak{E}_0^{\text{ho}} = \int_{\text{BZ}} \frac{d\mathbf{q}}{\Omega^{\text{BZ}}} \sum_\nu \frac{\omega_{\nu\mathbf{q}}}{2}$$

$$E^{\text{elph}} \equiv \lambda_0^4 \mathcal{E}^{(4),\text{elph}}$$

$$= \frac{1}{2} \iint_{\text{BZ}} \frac{d\mathbf{k}}{\Omega^{\text{BZ}}} \frac{d\mathbf{q}}{\Omega^{\text{BZ}}} \sum_n^{\text{occ}} \sum_{\kappa\alpha} \frac{1}{M_\kappa} \left\langle \frac{\partial \phi_{n\mathbf{k}}}{\partial R_{\kappa\alpha\mathbf{q}}} \left| \hat{\mathcal{P}}_{\perp, \text{occ}, \mathbf{k}}^{\text{BO}} \right| \frac{\partial \phi_{n\mathbf{k}}}{\partial R_{\kappa\alpha\mathbf{q}}} \right\rangle.$$

$$= \frac{1}{2} \iint_{\text{BZ}} \frac{d\mathbf{k}}{\Omega^{\text{BZ}}} \frac{d\mathbf{q}}{\Omega^{\text{BZ}}} \sum_n^{\text{occ}} \sum_m^{\text{unocc}} \sum_\nu \frac{2\omega_{\nu\mathbf{q}} |g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2}{(\varepsilon_{m\mathbf{k}+\mathbf{q}} - \varepsilon_{n\mathbf{k}})^2},$$



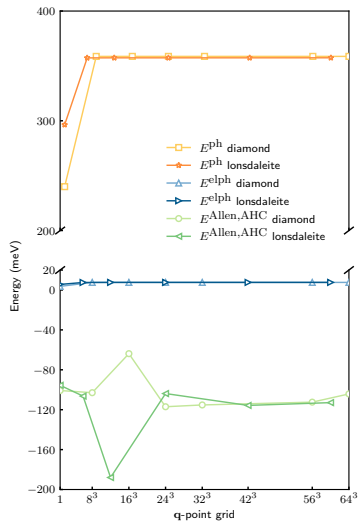
$|g_{mn\nu}(\mathbf{k}, \mathbf{q})|^2$ is EPC $\rightarrow$ the Born term is the EPC to total energy.

Verification on diamond with finite difference, $\mathbf{q} = \Gamma$

Total energy per primitive cell (2 atoms) + $8 \times 8 \times 8$ **k**-grid at 0 K.

	Evaluation	Energy (meV)
ω_{Γ}	DFPT	159.989
ω_{Γ}	FD	160.002
$\Delta \epsilon_{\Gamma}$	FD	27.695
$\Delta \epsilon_{\Gamma}^{\text{NR1A}}$	FD	3.894
$\Delta \epsilon_{\Gamma} - \Delta \epsilon_{\Gamma}^{\text{NR1A}}$	FD	23.801
$\Delta \epsilon_{\Gamma}^{\text{AHC}}$	DFPT	23.806
E^{BO}	DFT	-327559.404
E^{ph}	DFPT	240.040
E^{ph}	FD	240.023
E^{ph}	FD	240.003
E^{elph}	DFPT	3.639
E^{elph}	DFPT	3.638

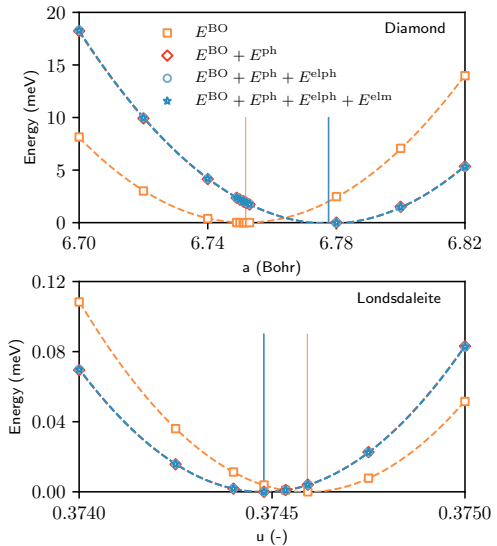
Momentum q convergence



S. Poncé and X. Gonze, arXiv:2512.04897 (2025)

Diamond and lonsdaleite (4 atoms = hexagonal diamond)

Total energy term	diamond (meV)	lonsdaleite (meV)
E^{BO}	-327559.216	-327505.244
E^{ph}	358.685	357.381
E^{elph}	7.600	7.608
E_1^{elph}	0.509	0.517
E_2^{elph}	1.510	1.476
E_3^{elph}	2.516	2.419
E_4^{elph}	3.066	3.197
$E^{\text{BO}} + E^{\text{ph}} + E^{\text{elph}}$	-327192.931	-327140.255
Relative difference	-52.676	0
ΔE^{QH}	-2.103	-



S. Poncé and X. Gonze, arXiv:2512.04897 (2025)

Implementation inside Abinit

Released in Abinit v10.4.5

See new tests v10/t61-t65

Total energy information automatically printed when
optdriver 7

eph_task 4

eph_ahc_type 0 ! Adiabatic, required

Added data inside XXXo_DS1_SIGEPH.nc



```
=====
Contributions to total energies (in meV)
  See Table II of S. Ponce and X. Gonze, arXiv:2512.04897
  for additional information.
=====
```

```
Temperature =      0.000000 K
E^(B0)  =    -325151.619138
E^(ph)  =      337.801155
E^(elph) =       6.103468
```

```
== END DATASET(S) =====
=====
```



Important limitation:

If coarse $\mathbf{k}/\mathbf{q}$ grids are same as fine $\rightarrow$ no problem

If you used perturbed potential interpolation $\rightarrow E^{\text{elph}}$ will be wrong because of wrong weights (documented in the code).

You need to read `XXXo_DS1_SIGEPH.nc` file and apply the correct $\mathbf{k}$ -point weights.

Acknowledgments

- “ALPS” online meeting !
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